STAT 410

Homework #11

Spring 2024
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(due Friday, April 26, by 11:59 p.m. CDT)

No credit will be given without supporting work.

7. Let $\beta > 0$ be a population parameter, and let $X_1, X_2, \ldots, X_n$ be a random sample (i.i.d.) from a distribution with probability density function

$$f(x;\beta) = \frac{1}{2\beta^2} e^{-\sqrt{x}/\beta}, \qquad x > 0, \qquad \text{zero otherwise.}$$

Recall: $W = \sqrt{X}$ has a Gamma$(\alpha = 2,\ \theta = \beta)$ distribution.

The maximum likelihood estimator for β,

$$\hat{\beta} = \frac{1}{2n} \sum_{i=1}^{n} \sqrt{X_i} = \frac{1}{2} \overline{W},$$

is an unbiased estimator of β.

v) Find a sufficient statistic for β.

w) Find the Fisher information $I(\beta)$.

(After you are done with part (w), glance back at part (o) on Homework #10.)

x) Is $\hat{\beta}$ an efficient estimator of β?

If $\hat{\beta}$ is not an efficient estimator of β, find its efficiency.

Is $\hat{\beta}$ an asymptotically efficient estimator of β?

① Find $\mathrm{Var}(\hat{\beta})$. ("Hint": We may or may not have done this already.)

② Find the Rao-Cramér lower bound.

③ Is $\hat{\beta}$ an efficient estimator of β? Does $\mathrm{Var}(\hat{\beta})$ attain the R.C.L.B.?

If $\hat{\beta}$ is not efficient, find its efficiency. Does efficiency approach 1 as $n \to \infty$?

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8. Let $\delta > 0$ be a population parameter, and let $X_1, X_2, \ldots, X_n$ be a random sample (i.i.d.) from a distribution with probability density function

$$f(x;\delta) = \frac{\delta}{(1+x)^{\delta+1}}, \qquad x > 0, \qquad \text{zero otherwise.}$$

Recall: $W = \ln(1+X)$ has a $\text{Gamma}(\alpha = 1, \theta = \frac{1}{\delta})$ distribution.

$$\hat{\hat{\delta}} = \frac{n-1}{\sum_{i=1}^{n} \ln(1+X_i)} \text{ is an unbiased estimator of } \delta.$$

n) Find a sufficient statistic for δ.

o) Find the Fisher information $I(\delta)$.

(After you are done with part (o), glance back at part (l) on Homework #10.)

p) Is $\hat{\hat{\delta}}$ an efficient estimator of δ?

If $\hat{\hat{\delta}}$ is not an efficient estimator of δ, find its efficiency.

Is $\hat{\hat{\delta}}$ an asymptotically efficient estimator of δ?

① Find $\text{Var}(\hat{\hat{\delta}})$. ("Hint": You may or may not find it helpful to revisit part (g) on Homework #09.)

② Find the Rao-Cramér lower bound.

③ Is $\hat{\hat{\delta}}$ an efficient estimator of δ? Does $\text{Var}(\hat{\hat{\delta}})$ attain the R.C.L.B.?

If $\hat{\hat{\delta}}$ is not efficient, find its efficiency. Does efficiency approach 1 as $n \to \infty$?

10. Assume that the population is normally distributed with unknown mean μ and standard deviation $\sigma = 18$. We plan to use a sample of size $n = 9$ to test $H_0 : \mu = 70$ vs. $H_1 : \mu > 70$ at a 5% level of significance.

a) If the sample mean is $\bar{x} = 79.24$, find the p-value of the test.

b) Find the Rejection Region for the test at $\alpha = 0.05$. That is, for which values of the sample mean $\bar{X}$ should we reject H_0, if a 5% level of significance is used?

c) Find the power of the test if the actual value of the population mean is $\mu = 75.31$, and a 5% level of significance is used.

11. Assume that the population is normally distributed with unknown mean μ and standard deviation $\sigma = 18$. We plan to use a sample of size $n = 9$ to test $H_0 : \mu = 70$ vs. $H_1 : \mu \neq 70$ at a 5% level of significance.

a) If the sample mean is $\bar{x} = 79.24$, find the p-value of the test.

b) Find the Rejection Region for the test at $\alpha = 0.05$. That is, for which values of the sample mean $\bar{X}$ should we reject H_0, if a 5% level of significance is used?

c) Find the power of the test if the actual value of the population mean is $\mu = 76.54$, and a 5% level of significance is used.